

2018 Ph H2 Q6

Section: Particles and Waves

Topic: Forces on Charged Particles

Summary: Electrons are accelerated between cathode and anode by a potential difference, then travel past (or are blocked by) a metal cross to strike a fluorescent screen. You must calculate work done, electron speed, suggest an observational change when voltage increases, and compare a tabletop model to a real particle accelerator.

(a)(i) Work done

Relationship: $W = QV$

$$Q = e = 1.60 \times 10^{-19} \text{ C}$$

$$V = 1.60 \times 10^3 \text{ V}$$

$$W = (1.60 \times 10^{-19})(1.60 \times 10^3) = 2.56 \times 10^{-16} \text{ J} \approx 2.6 \times 10^{-16} \text{ J (as required)}$$

(a)(ii) Electron speed at anode

From $\frac{1}{2}mv^2 = W$:

$$m = 9.11 \times 10^{-31} \text{ kg}$$

$$v = \sqrt{(2W/m)}$$

$$v = \sqrt{(2 \times 2.56 \times 10^{-16} / 9.11 \times 10^{-31})} = 2.38 \times 10^7 \text{ m/s}$$

(b) Change when voltage increases

Screen will be brighter / glow more intensely.

Justification: electrons gain more kinetic energy and/or more electrons per second reach the screen, increasing light output.

(c) Comparison of model to real accelerator

The model shows repeated acceleration at one point on a circular path, similar to the way particles in a synchrotron gain energy in stages. However, real accelerators such as the LHC use electric fields in radiofrequency cavities rather than a mechanical motor, operate in high vacuum, and accelerate subatomic particles to near-light speeds. The plastic block collision is analogous to a target or particle beam collision, but in reality the collisions involve very high energies and require sophisticated detectors to observe products.

Final answers

(a)(i) $2.6 \times 10^{-16} \text{ J}$

(a)(ii) $2.38 \times 10^7 \text{ m/s}$

(b) Brighter screen / increased glow

(c) Model shows basic repeated acceleration and collision idea, but real accelerators use electric fields, vacuum, and subatomic particles at relativistic speeds.

Revision tips

- For charged particle acceleration, $W = QV$ gives energy gained from a potential difference.
- Use $\frac{1}{2}mv^2 = W$ for non-relativistic speeds ($v \ll c$).
- Qualitative questions often want cause-effect links: say what changes, then why, in physics terms.
- When comparing models, identify similarities and differences in principles, scale, and technology.